- $A \otimes B$: 4 × 4
- $a \otimes b$: 4 × 1
- $Aa \otimes Bb$: 4 × 4

e) Perform the matrix multiplication on the left-hand side, resulting in a 4 × 1 column vector. Each row should be the sum of four separate terms.

f) Finally, verify that the resulting column vectors on the left and right sides are identical.

7.2 MATHEMATICAL INTERLUDE: OUTER PRODUCTS

Given a bra $\langle\varphi|$ and a ket $|\psi\rangle$, we can form the inner product $\langle\varphi|\psi\rangle$. As we've seen, the inner product is a complex number. However, there is another kind of product called the outer product, written

$$|\psi\rangle\langle\phi|.$$

The outer product is not a number; it is a linear operator. Let's consider what happens when $|\psi\rangle\langle\varphi|$ acts on another ket $|A\rangle$:

$$|\psi\rangle\langle\phi| \quad |A\rangle.$$

In these examples, we're using spacing instead of parentheses to show the grouping of operations. Remember that all operations with bras, kets, and linear operators are associative, which means we're allowed to group them any way we like, as long as we keep the same ordering from left to right.[3] The action of the outer product operator is very simple and can be defined as

$$|\psi\rangle\langle\phi| \quad |A\rangle \equiv |\psi\rangle \quad \langle\phi|A\rangle.$$